In-Blender Post-Processing, Robustness & Future Testing

Technical Progress Report (Phase 5 & 6)

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FOSSEE Internship – Progress Report

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April 12, 2026

1. Integrated STL Importer

- **Objective**: Allow users to import external CAD models (.stl) directly from the Blender sidebar — the standard entry point for any CFD workflow.
- **Implementation**:
  - New operator `OBJECT_OT_ImportSTL` with native Blender file browser (filtered to `*.stl`).
  - Auto-selects imported geometry and reports vertex/face count.
  - Full validation: empty files, missing files, wrong extension → red error box.

*STL Import Button in the Mesh Generation Panel

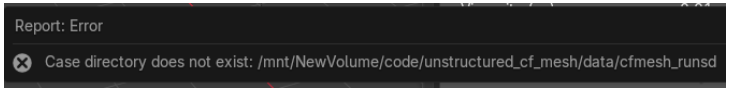
2. Comprehensive Input Validation (15+ Guards)

Week 4 Problem

Invalid directories, zero velocity, or wrong solver/turbulence combinations could silently crash OpenFOAM with cryptic terminal errors.

Week 5 Solution — Every operator now validates before execution:

- **GenerateCFMesh**: Path validation, cell size sanity (< 0.01 too small), empty geometry, write permissions.
- **RunSolver**: Zero velocity guard, viscosity > 0, solver-turbulence compatibility (icoFoam $\neq$ kOmegaSST).
- **All Operators**: Concurrent run guard prevents stacking processes.



*Red Alert Box for Errors

3. OpenFOAM 11 Solver Compatibility Fix

Critical Discovery

`simpleFoam` was **deprecated** in OpenFOAM 11 — it only prints a deprecation message and exits. The solver never actually ran!

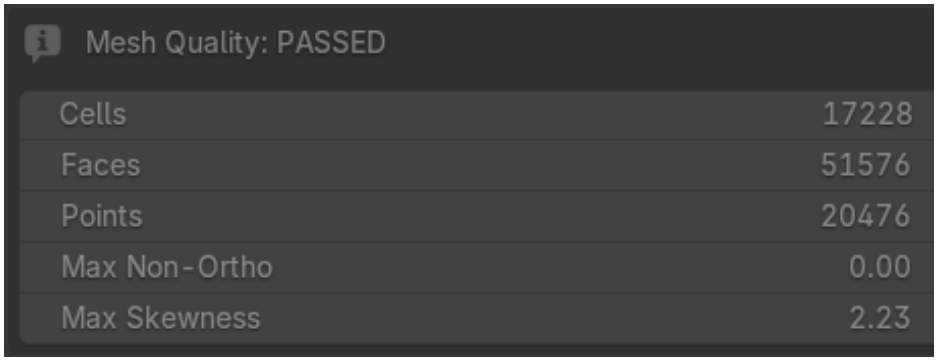
Resolution:

- Mapped `simpleFoam` → `foamRun -solver incompressibleFluid` in the backend.
- `icoFoam` still functions natively in OF11.
- This fix is transparent to the user — they still select "simpleFoam" in the UI dropdown but the correct OF11 command runs underneath.

UI Selection	OF10 Command	OF11 Command
icoFoam	`icoFoam`	`icoFoam`
simpleFoam	`simpleFoam`	`foamRun -solver incompressibleFluid`

4. In-Blender Mesh Quality Inspector

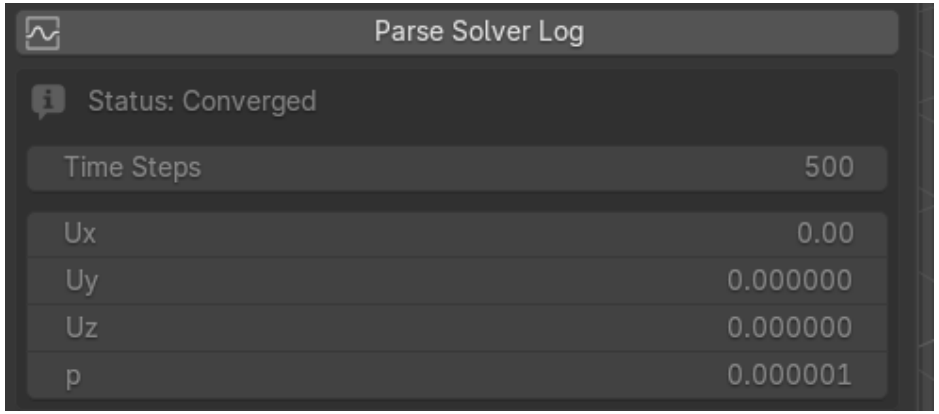
- **Objective**: Let users verify mesh quality without leaving Blender.
- Runs OpenFOAM's `checkMesh` and parses the output:
  - Cell count, Face count, Point count.
  - Max Non-Orthogonality (ideal < 65).
  - Max Skewness (ideal < 4).
  - PASSED / FAILED verdict.



*Mesh Quality: 17,228 cells, PASSED

5. Solver Residual Monitor

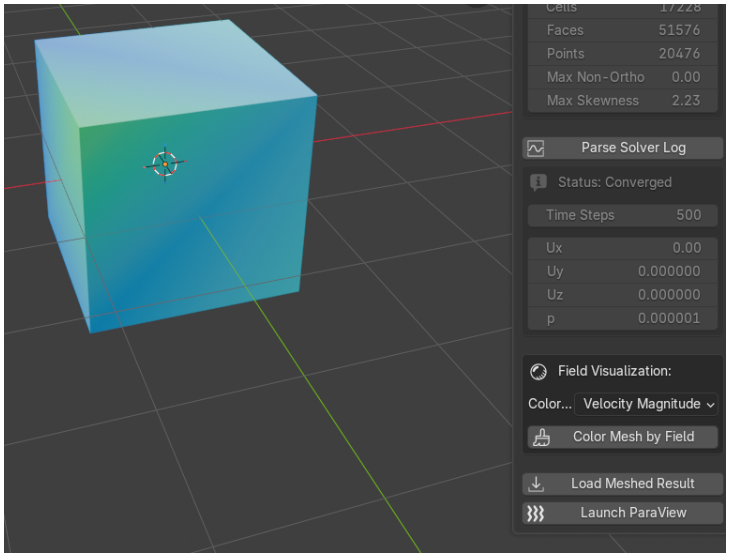
- Parses the solver's live output log to extract final residual values for each field:
  - Velocity components: U_x, U_y, U_z
  - Pressure: p
  - Turbulence: k, ω/ϵ
- Counts time steps and determines convergence status:
  - **Converged**: max residual $< 10^{-4}$
  - **In Progress**: residuals still decreasing
  - **Diverged**: max residual > 1.0



*Solver Residual Display in the Post-Processing Panel

6. Field Visualization — Color Mesh by Pressure / Velocity

- **The Key Feature**: Visualize CFD results *directly in Blender's viewport* without needing ParaView.
- Reads OpenFOAM field data from the latest time step (17,228 values for pressure).
- Applies a **Jet Colormap** (Blue → Cyan → Green → Yellow → Red) as vertex colors on the mesh.
- Auto-switches viewport to Material Preview for instant visual feedback.



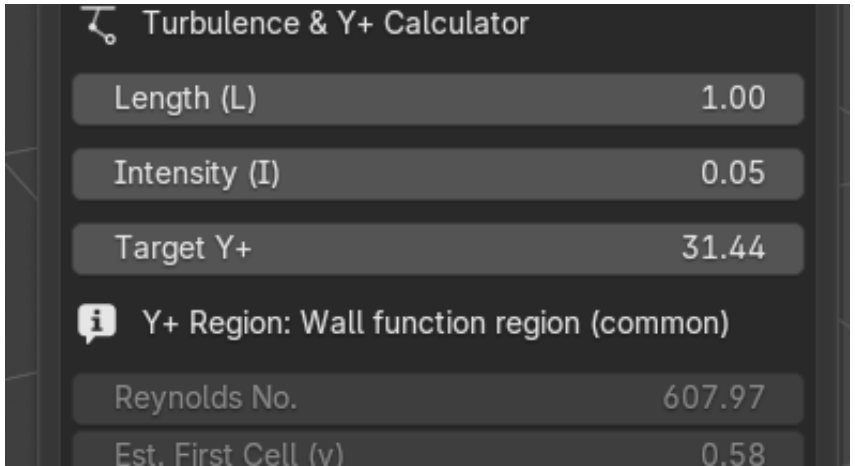
*Pressure Colormap on Mesh

Supported fields: `Pressure (p)` and `Velocity Magnitude (|U|)`

7. Live Y+ & Turbulence Calculator

- Real-time sidebar calculator that computes turbulence parameters from physical inputs:

Input	**Computed Output**
Velocity, Length, Viscosity	Reynolds Number (Re)
Target Y^+	First Cell Height (y)
Turbulent Intensity (I)	$k = \frac{3}{2}(U \cdot I)^2$
	$\epsilon = C_\mu^{3/4} k^{3/2} / l$
	$\omega = k^{1/2} / (C_\mu^{1/4} \cdot l)$



8. Codebase Modularization & UX Enhancements

- **Monolith to Modules**: Split monolithic scripts into `ui.py`, `operators.py`, `properties.py` for clean architecture and better maintainability.
- **Asynchronous Parsing**: OpenFOAM solver logs and `checkMesh` results are parsed smoothly.
- **Remodeled Interface**: Reorganized the add-on panels for a logical top-to-bottom CFD workflow: Geometry → Mesh → Physics → Post-Processing.

9. Upcoming Work (Week 6) — Rigorous Testing

- **Objective**: Validate the add-on across diverse scenarios to ensure it is production-ready.
- **Planned Test Scenarios**:
  - **Geometries**: Testing STL importer and `snappyHexMesh` with multi-region and complex curved geometries.
  - **Physics Models**: Validating high Reynolds number simulations against the computed $k - \omega$ SST values.
  - **Scale Benchmarks**: Performance testing vertex coloring with field data on meshes exceeding 500,000 cells.
  - **Edge Cases**: Handling corrupted STL files, abruptly terminated solvers, and low disk space gracefully.

Progress Summary

Feature/Milestone	Outcome/Benefit
STL Importer	Import CAD models directly into Blender
15+ Validation Guards	No silent crashes, explicit error reporting
OF11 Solver Fix	simpleFoam → foamRun compatibility
Residual Monitor	Track solver convergence in the UI
Color by Field	Visualize pressure/velocity in viewport
Codebase Modularization	Refactored to ui, operators, properties
Upcoming week	Production-grade scenarios & benchmarks

Addon Version

Core features in `cfmesh_tools_2.3.zip`. Entering Week 6 Testing Phase.

Thank You!

End of Week 5 & 6 Report